

PROBABILITY AND RANDOM VARIABLES-1

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Let A and B be disjoint events and assume that $P_r(A) = P_r(B)$ and $P_r(A \cup B) = 0.8$. Then $P_r(A) =$

Select one:

- ☐ a. 0.1
- ☐ b. 0.4
- ☐ c. 0.2
- ☒ d. 0.3

d

ABILITY AND RANDOM VARIABLES-1

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For a random variable X , suppose that $E(X^2) = 3$, then $E(2X^2 - 1) =$

Select one:

☐ a. 3

☐ b. 7

☐ c. 5

☐ d. 1

C

5

answered

out of 1.00

question

For a discrete random variable X , which of the following can be a c.d.f

Select one:

☒ a. $F(X) = \begin{cases} 0 & , X < 0 \\ \frac{1}{3} & , 0 \leq X < 2 \\ 1.1 & , X \geq 2 \end{cases}$

☐ b. $F(X) = \begin{cases} 0.1 & , X < 0 \\ \frac{1}{5} & , 0 \leq X < 2 \\ 1 & , X \geq 2 \end{cases}$

☐ c. $F(X) = \begin{cases} 0 & , X < 0 \\ \frac{1}{3} & , 0 \leq X < 1 \\ 1 & , X \geq 1 \end{cases}$

☐ d. $F(X) = \begin{cases} 0.1 & , X < 0 \\ \frac{1}{6} & , 0 \leq X < 2 \\ 1.1 & , X \geq 2 \end{cases}$

c

Let X be a continuous random variable with pdf $f(X) = e^{-\lambda X}$ when $X > 0$ and zero otherwise. The constant λ equals

Select one:

☐ a 0.6

☐ b 1

☐ c 0.5

☐ d 0.3

v

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Let X be a continuous random variable. If $P_r(a < X < b) = 0.1$. Then $P_r(a \leq X < b) =$

Select one:

- ☒ a. 0.2
- ☐ b. 0.1
- ☐ c. 0.3
- ☐ d. 0.4

b

Which of the following is **always** true

Select one:

- ☐ a. The variance of any random variable cannot be negative
- ☐ b. The correlation of two random variables can equal 2
- ☐ c. The covariance of any two random variables cannot be negative
- ☐ d. The correlation of any two random variables cannot be negative

a

PROBABILITY AND RANDOM VARIABLES-1

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Let X be some random variable with $Var(X) = 1$, then $Var(-1 - X) =$

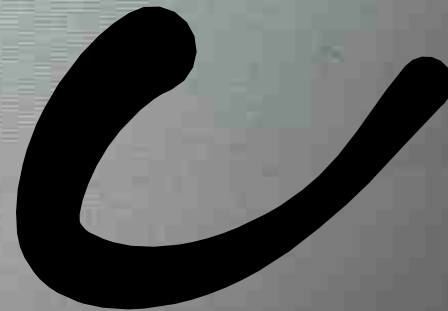
Select one:

☒ a. 9

☐ b. 16

☐ c. 1

☐ d. 4



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For a continuous random variable X which of the following is **wrong** about the c.d.f $F(X)$

Select one:

- ☐ a. $\lim_{t \rightarrow \infty} F(X) = 1$
- ☐ b. $\lim_{t \rightarrow -\infty} F(X) = 0$
- ☐ c. $F(X)$ is increasing
- ☐ d. $F(X)$ is a step function

d

Randomly we draw 3 balls from a box that contains 3 red balls and 2 black balls. The probability of choosing 2 black balls is

Select one:

☒ a. $\frac{\binom{3}{3}\binom{2}{0}}{\binom{5}{3}}$

☐ b. $\frac{\binom{3}{3}\binom{2}{1}}{\binom{5}{3}}$

☐ c. $\frac{\binom{3}{1}\binom{2}{2}}{\binom{5}{3}}$

☐ d. $\frac{\binom{3}{3}\binom{2}{1}}{\binom{5}{3}}$

b

A continuous random variable X has c.d.f $F(x)$. For some numbers a and b with $a < b$ if $P_r(a < X < b) + F(a) = 0.3$ Then $F(b) =$

Select one:

☐ a. 0.1

☐ b. 0.3

☐ c. 0.2

☐ d. 0.4

b

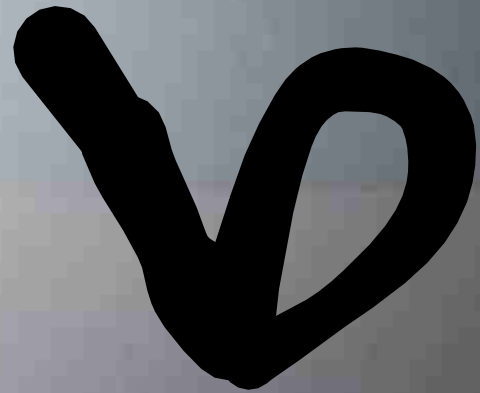
ABILITY AND RANDOM VARIABLES-1

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Two independent random variables X and Y with $E(X) = 0.1$ and $E(Y) = 0.4$. Then $E(XY)$

Select one:

- ☒ a. 0.02
- ☐ b. 0.04
- ☐ c. 0.05
- ☐ d. 0.03



Two events A and B with $P_r(\bar{B}) = 0.4$ and $P_r(A - B) = 0.1$. Then $P_r(A|B) =$

Select one:

- ☐ a. $\frac{1}{4}$
- ☐ b. 1
- ☐ c. $\frac{1}{3}$
- ☐ d. $\frac{1}{2}$

c

Question 13

Not yet answered

Marked out of 2.00

Flag question

Two continuous and independent random variables X and Y with pdfs

$$f_X(x) = 1 \quad \text{when } 0 < x < 1 \quad \text{and zero otherwise}$$

$$f_Y(y) = e^{-y} \quad \text{when } y > 0 \quad \text{and zero otherwise}$$

Then the joint pdf is $f(x, y) =$

Select one:

☐ a.

$$\frac{e^{-y}}{4}$$

☐ b.

$$\frac{e^{-y}}{2}$$

☐ c.

$$e^{-y}$$

☐ d.

$$\frac{e^{-y}}{3}$$

C

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For random variables X and Y , we have $E(X) = E(Y) = 1$ and $Cov(X, Y) = -4$. Then $E(XY) =$

Select one:

☐ a. -4

☐ b. -2

☐ c. -3

☐ d. -1

C

For an event A , we have $P_r(\emptyset | A) + 2 P_r(\Omega | A) =$

Select one:

☐ a. 4

☐ b. 5

☐ c. 3

☐ d. 2

d

ABILITY AND RANDOM VARIABLES-1

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Let X and Y be some random variables with joint pdf $f(x, y) = \begin{cases} \frac{1}{2} & , 0 \leq x \leq 2, 0 \leq y \leq 1 \\ 0 & , \text{Otherwise} \end{cases}$ The marginal pdf with respect to Y is

Select one:

☐ a. $\frac{x}{2}$

☐ b. 1

☐ c. $\frac{1}{2}$

☐ d. $\frac{1}{x}$

b

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Let X and Y be some random variables with joint pdf $f(x, y) = \begin{cases} \frac{1}{2} & , 0 \leq x \leq 2, 0 \leq y \leq 1 \\ 0 & , \text{Otherwise} \end{cases}$. The marginal pdf with respect to Y is $\square$

Select one:

- ☒ a. $\frac{X}{2}$
- ☐ b. 1
- ☐ c. $\frac{1}{2}$
- ☐ d. $\frac{1}{X}$



Next p

ABILITY AND RANDOM VARIABLES-1

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For random variables X and Y assume that $Cov(X, Y) = 2 Corr(3X, 2 + Y)$. Then $Var(X)Var(Y) =$

Select one:

- ☐ a. 4
- ☐ b. 16
- ☐ c. 9
- ☐ d. 25

a

ABILITY AND RANDOM VARIABLES-1

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Let X and Y be some random variables with range $R_{XY} = \{(0,0), (0,1), (1,0), (1,1)\}$. If $f(0,0) = f(0,1) = f(1,0) = f(1,1)$. Then $f(1,1) =$

Select one:

☒ a. $\frac{1}{7}$

☐ b. $\frac{1}{4}$

☐ c. $\frac{1}{6}$

☐ d. $\frac{1}{5}$

b

For independent random variables X and Y , we have $Cov(1 - X, 2 + \pi Y) =$

Select one:

☒ a. $-\pi$

☐ b. 1

☐ c. 0

☐ d. π

